

**COMPREHENSIVE CANCER CENTER**  
Leukemia Clinical Research Unit

April 15, 2024

Dr. Margaret Foster  
Community Oncology Associates  
4500 Riverside Drive  
Columbus, OH 43210

**RE: Angela Turner (DOB: 01/15/1960, MRN AML048) - Enrollment in Maintenance Therapy Trial**

Dear Dr. Foster,

I am writing to provide you with a comprehensive update on our mutual patient, Mrs. Angela Turner, who has recently enrolled in our institutional clinical trial evaluating maintenance therapy strategies for patients with intermediate-risk AML in first complete remission. This letter will detail her disease course, current status, and the specifics of the clinical trial protocol.

**COMPREHENSIVE DISEASE HISTORY:**

Mrs. Turner, a 64-year-old retired high school biology teacher, initially presented to your office in March 2023 with a 6-week history of progressive fatigue, easy bruising, and gingival bleeding while flossing. Your astute clinical assessment and prompt laboratory evaluation revealed concerning findings that led to her immediate referral to our center.

*Initial Presentation (04/03/2023):*

- WBC: 34,600/ $\mu$ L with 62% circulating blasts
- Hemoglobin: 7.2 g/dL
- Platelets: 18,000/ $\mu$ L
- Comprehensive metabolic panel: Notable for LDH 1,234 U/L, otherwise unremarkable
- Coagulation studies: PT 14.2 seconds, INR 1.2, PTT 32 seconds, Fibrinogen 145 mg/dL (mild DIC)

*Diagnostic Evaluation:* Her bone marrow biopsy performed on 04/04/2023 revealed:

- Morphology: 78% myeloblasts with minimal differentiation
- Immunophenotype: CD34+, CD117+, CD13+, CD33+, HLA-DR+, CD11b partially positive
- Cytogenetics: t(9;11)(p21.3;q23.3) - KMT2A (formerly MLL) rearrangement in 20/20 metaphases
- Molecular panel: KMT2A-MLLT3 fusion confirmed by FISH and RT-PCR, wild-type for FLT3, NPM1, CEBPA, IDH1/2, and TP53
- Classification: AML with t(9;11)(p21.3;q23.3); MLLT3-KMT2A, intermediate risk per ELN 2022

## TREATMENT COURSE TO DATE:

*Induction Therapy (April 11-17, 2023):* Mrs. Turner received standard 7+3 induction with:

- Cytarabine 100 mg/m<sup>2</sup> continuous infusion days 1-7 (total dose per day: 168 mg based on BSA 1.68 m<sup>2</sup>)
- Daunorubicin 90 mg/m<sup>2</sup> IV days 1, 2, and 3 (dose intensified given age <65 and good performance status)

Induction complications included:

- Febrile neutropenia day 8-21 with *Enterococcus faecium* bacteremia (successfully treated with linezolid)
- Grade 2 mucositis managed with magic mouthwash and low-dose morphine PCA
- Mild tumor lysis syndrome (peak uric acid 9.8) managed with allopurinol and hydration
- Transfusion requirements: 8 units pRBCs, 12 units platelets over 4 weeks

*Day 14 Assessment:* Hypocellular marrow with 3% residual blasts *Day 28*

*Assessment:* Complete remission achieved - 2% blasts, ANC 1,100, platelets 95,000 (CRi)

*Consolidation Therapy (May 2023 - October 2023):* Given her KMT2A rearrangement and intermediate risk, we proceeded with 4 cycles of HiDAC consolidation:

Cycle 1 (May 22-24, 2023):

- Cytarabine 3 g/m<sup>2</sup> IV over 3 hours q12h on days 1, 3, and 5
- Complicated by grade 2 cerebellar toxicity (ataxia, nystagmus) leading to dose reduction

Cycles 2-4 (July, August, October 2023):

- Cytarabine dose reduced to 2 g/m<sup>2</sup> due to neurotoxicity
- Each cycle required 5-7 day hospitalization for febrile neutropenia
- Cumulative toxicities: Mild hearing loss (high frequency), peripheral neuropathy grade 1

*MRD Assessment Post-Consolidation (November 2023):*

- Multiparameter flow cytometry: MRD negative (<0.01%)
- RT-PCR for KMT2A-MLLT3: Transcript levels decreased by 3.5 logs from diagnosis
- Bone marrow morphology: Normocellular with trilineage hematopoiesis, no excess blasts

**CURRENT STATUS (11 Months from Diagnosis):**

Mrs. Turner remains in complete morphologic and molecular remission. Her most recent evaluation (04/01/2024) showed:

*Performance Status:* ECOG 0 - fully active, teaching part-time as substitute teacher

*Physical Examination:*

- Vital signs stable, weight recovered to baseline (145 lbs)
- No lymphadenopathy or organomegaly
- Neurologic exam normal except mild decreased sensation in fingertips

*Laboratory Studies (04/08/2024):*

- CBC: WBC 4,200 (ANC 2,100, ALC 1,400), Hemoglobin 12.8 g/dL, Platelets 189,000
- Chemistry: All values within normal limits including creatinine 0.8 mg/dL
- Hepatic panel: Normal
- KMT2A-MLLT3 transcript: Undetectable by sensitive RT-PCR (sensitivity  $10^{-5}$ )

## **CLINICAL TRIAL ENROLLMENT:**

Given her excellent response but intermediate-risk disease with KMT2A rearrangement (historically 40-50% relapse rate), Mrs. Turner has enrolled in our institutional protocol:

*Trial Title:* "A Randomized Phase II Study of Oral Azacitidine versus Observation for Maintenance Therapy in Patients with Intermediate-Risk AML in CR1" (Protocol #AML-MAINT-2024)

*Rationale:* While patients with favorable-risk AML often achieve cure with chemotherapy alone, intermediate-risk patients have variable outcomes. KMT2A-rearranged AML, while classified as intermediate-risk, has a substantial relapse rate. This trial evaluates whether maintenance therapy with oral azacitidine can reduce relapse risk.

*Trial Design:*

- Randomization: 1:1 to oral azacitidine (CC-486) versus observation
- Stratification factors: Age (<60 vs ≥60), MRD status, cytogenetic risk group
- Mrs. Turner randomized to: ORAL AZACITIDINE ARM

*Treatment Plan:*

- Oral azacitidine 300 mg daily days 1-14 of each 28-day cycle
- Continue for 12 cycles or until relapse/intolerance
- Dose modifications for cytopenias or GI toxicity

*Monitoring on Trial:*

- CBC weekly for cycle 1, then biweekly

- Bone marrow biopsy with MRD assessment every 3 months
- KMT2A-MLLT3 RT-PCR monthly
- Quality of life assessments using FACT-Leu questionnaire
- Correlative studies: Serial samples for clonal evolution analysis

*First Cycle Experience (Started 04/12/2024):*

- Day 1-3: Mild nausea managed with ondansetron
- Day 4-14: Tolerated well, no significant toxicity
- Current labs (day 14): WBC 3,100, Hgb 11.9, Plt 156,000

**PROGNOSIS AND FUTURE CONSIDERATIONS:**

For patients with t(9;11) AML achieving CR1, published literature suggests:

- 5-year overall survival: 45-55%
- 5-year relapse-free survival: 35-45%
- Median time to relapse (if occurs): 11-14 months

The addition of maintenance therapy potentially could improve these outcomes by 10-15% based on preliminary data from similar trials. We are optimistic about Mrs. Turner's prognosis given her:

- Deep molecular remission (undetectable MRD)
- Excellent performance status
- Good tolerance of therapy
- Strong psychosocial support system

**COORDINATION OF CARE:**

While Mrs. Turner is on trial, we will:

- See her in our clinic every 2 weeks for the first 2 cycles, then monthly
- Send you clinic notes after each visit
- Contact you immediately with any significant changes
- Coordinate any local laboratory draws if needed

Please continue to manage her routine health maintenance including:

- Mammography (due next month)
- Colonoscopy (completed 2022, next due 2032)
- Bone density scan (recommended given chemotherapy exposure)
- Annual influenza vaccination (coordinate timing with us)

**PSYCHOSOCIAL CONSIDERATIONS:**

Mrs. Turner has demonstrated remarkable resilience throughout her treatment. She actively participates in our AML support group and has become a peer mentor for newly diagnosed patients. Her husband of 38 years has been exceptionally supportive, attending all appointments and managing her complex medication

schedule. Their two adult children (both healthcare workers) have been instrumental in her care.

She has expressed some anxiety about relapse risk, which is being addressed through:

- Monthly sessions with our psycho-oncologist
- Mindfulness-based stress reduction classes
- Participation in our survivorship program

Thank you for your continued partnership in Mrs. Turner's care. Her positive outcome thus far reflects the importance of early recognition and prompt referral. Please don't hesitate to contact me directly at 555-LEUK (5385) or via secure email if you have any questions or concerns.

With warm regards,

Jennifer M. Anderson, MD, PhD  
Associate Professor of Medicine  
Director, AML Clinical Trials Program  
Comprehensive Cancer Center

cc: Angela Turner  
Clinical Research Coordinator  
Trial Data Management Office